

Caesar Cypher

Problem-1: Basic Caesar Encryption (Uppercase Only)

Write a Python program that takes:

1. A plaintext string consisting only of uppercase English letters (A–Z)
2. An integer shift value k

and outputs the encrypted ciphertext using the Caesar Cipher rule:

Input Example

HELLO

3

Output Example

KHOOR

Assume:

- No lowercase letters
- No spaces or special characters

Problem-2: Preserve Non-Alphabet Characters

Modify your encryption program so that:

- Uppercase and lowercase letters are encrypted
- Digits, spaces, and punctuation remain unchanged

Input

Hello, World! 123

Shift = 5

Output

Mjqqt, Btwqi! 123

Problem-3: Caesar Cipher Menu-Driven Program

Write a program that shows this menu:

1. Encrypt a message
 2. Decrypt a message
 3. Exit
- The user chooses an option
 - Inputs message and key
 - Program displays the result
 - Continues until Exit is chosen

Problem-4: Brute-Force Attack (Cryptanalysis)

Write a Python program that:

- Takes only a ciphertext
- Tries all 26 possible shifts
- Prints every possible decrypted message

Input

ciphertext = "KHOOR"

Output (partial)

Shift 0: KHOOR

Shift 1: JGNNQ

Shift 2: IFMMP

Shift 3: HELLO

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Use this to show why Caesar cipher is insecure.

Problem-5: Negative Shift Handling

Extend your program so the key k may be:

- Positive
- Zero
- Negative

Example:

Input

HELLO

-3

Output

EBIL